SULIT

SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	Loan Management Module
System Name	KADA ESERVE
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SECTION A (SYSTEM'S OVERVIEW)

1.1Overall Description

The Loan Management Module of KADA-eServe revolutionizes KADA's loan application process by transitioning from a manual, paper-based system to a fully digitalized and database-driven system. This module focuses on simplifying and improving the efficiency of all loan-related processes, simultaneously reducing human error and providing a seamless user experience for KADA's staff, members, and clerks.

The system leverages a robust, structured database designed to manage all loan-related processes with precision. The database ensures data consistency, easy retrieval, data management, secure storage, and high scalability. By proposing this module, roles are assigned before member registration, clearly distinguishing approved members who can apply for loans. Members can submit loan applications, track statuses, and manage repayment details in real time, while clerks can efficiently validate and update loan applications. Furthermore, the module is designed to integrate with other KADA-eServe components, such as membership management and reporting systems, for streamlined operations. Additionally, the uniqueness of this module lies in the fact that members do not need to re-enter their information when applying for a loan. Since their details are already captured during the member registration process, the system automatically populates the application fields. This is made possible through the structured database design and the application of normalization process, which ensures data consistency and reduces data redundancy.

Group Members' Name	Matric No	Module
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1.2Project Weaknesses or Improvement

The current loan application system's weaknesses are a fully manual process, leading to fragmented data storage and higher chances of human error, duplication, delays, as well as data loss, as it only stores via form. The absence of a centralized database means that all loan-related information is stored in physical records, making it difficult to track, retrieve, and update data efficiently. The manual nature of the process increases the processing time, making it hard for KADA to manage an increasing amount of loan applications productively.

Therefore, improvements are needed to tackle those weaknesses and limitations. The proposed system, KADA-eServe, introduces a centralized digital database that integrates all loan-related data in a secure and accessible location. These improvements have significantly improved data integrity, reducing the risk of errors and inconsistencies. By digitalizing the process, the system will enable a faster loan application process with real-time updates. Additionally, the handling of increasing volumes of loan applications will be facilitated by the database structure, supporting future growth and ensuring a scalable loan management module.

1.3Proposed Module

The proposed Loan Management Module addresses the inefficiencies of the existing manual, paper-based system by introducing a fully digitalized system. It centralizes data storage, automates workflows, and minimizes errors, enabling a seamless loan application and management process.

The proposed module will include:

1. **Member Features**

   The proposed system provides enhanced functionality for members, such as:

   - **Digital Loan Application:** Members can submit loan applications online with pre-filled personal details to reduce manual input.
   - **Real-Time Status Tracking:** Members can track the progress of their loan applications through the status interface.
   - **Automated Repayment Management Details:** Loan repayment schedules and balances are displayed clearly, allowing members to make payments and monitor their financial obligations in real time.
   - **Document Export:** Members can export loan-related documents, including applications and repayment statements, in PDF format for easy record-keeping.

2. **Admin Features**

   Admin, also known as clerk. The features are designed to simplify loan management processes, ensure smooth system operations, and minimize admin involvement in member notifications:

   - **Loan Review and Approval:** Admins also validate loan applications, approve or reject them, and provide justifications.
   - **Automated Email Notifications:** Email notifications are sent to members upon application status changes.
   - **Dividend Rate Update:** Admins can adjust the dividend rate, and the updated values are reflected in loan calculations immediately.
   - **Transaction Update:** Admins can log repayments directly into the system for accurate record-keeping.

3. **Database Integration**

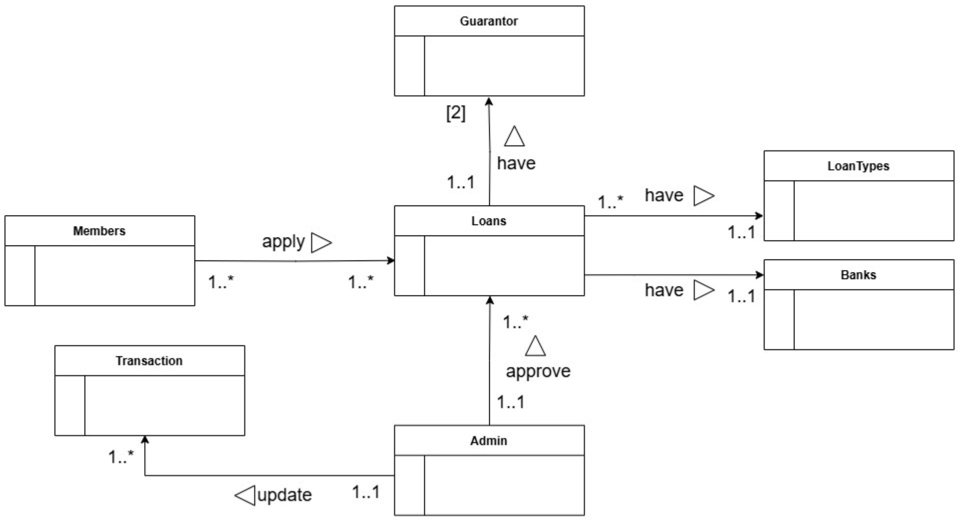
   The database plays the main role in the digital system, enabling advanced functionalities:

   - **Centralized Data Storage:** The unified database ensures all loan-related information is stored securely together and is easily accessible for processing.
   - **Pre-Filled Forms:** The database automatically populates member details during loan application to save time.

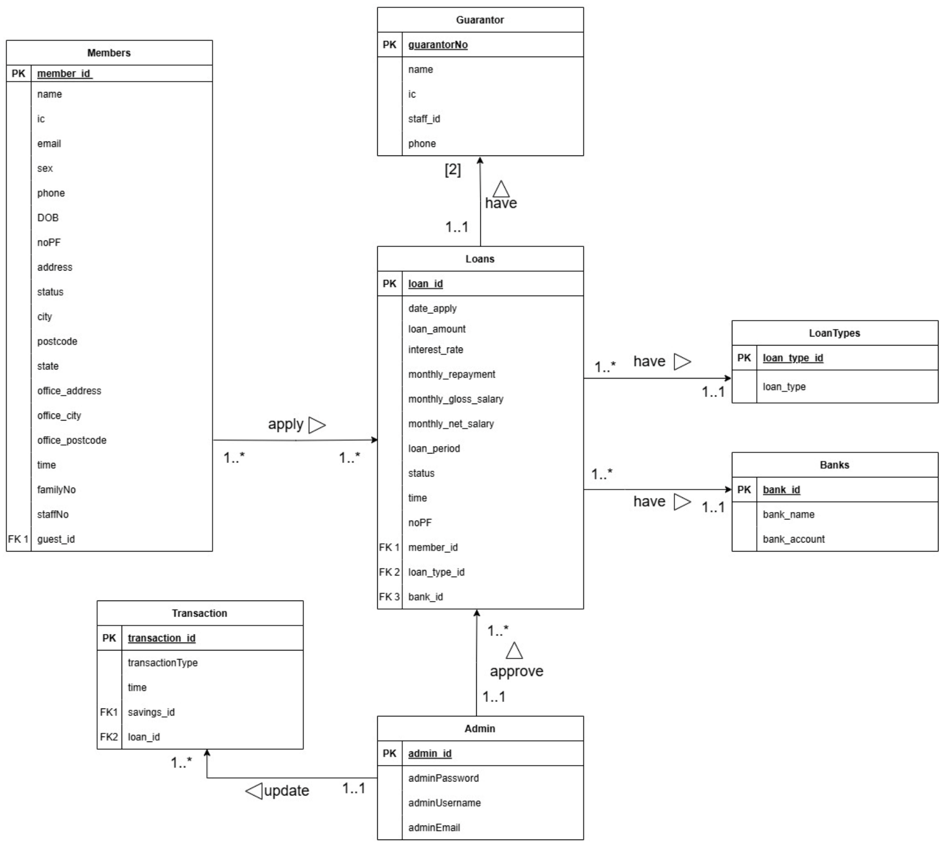
- **Normalization and Data Integrity:** The system reduces data redundancy and guarantee consistency across all records.
- **Real-Time Analytics and Reporting:** Both members and admins can generate reports and retrieve all data from the database.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

2.3.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
Member	General term describing registered and approved members using KADA-eServe	Registered staff	Occurs when a user (KADA staff) registers in the system
Admin	General term describing clerks that manage KADA-eServe	Clerk, Administrator	Occurs when system is built and require an administrator to manage and monitor system operations
Loans	General term describing loan applications made by members	Loan application details	Occurs when a member applies for a loan.
LoanTypes	General term describing different types of loans offered by KADA	Loan categories	Occurs when a member applies loan, specifying type of loan they required and when admin update transaction
Banks	General term describing bank associated with members' loan applications	Bank reference	Occurs when a member applies for a loan.
Guarantor	General term describing KADA staff who guarantee other members' loan applications	Financial guarantor	Occurs when a member provides two guarantors during loan application.
Transaction	General term describing all transaction record key-in by admin	Deposit amount, loan repayment	Occurs when admin key-in the amount of deposit or loan repayment for each member

2.3.2 Identifying Relationship Types

Entity name	Multiplicity	Relationship	Entity Name	Multiplicity
Member	1..*	apply	Loan	1..*
Admin	1..1	approve	Loan	1..*
	1..1	update	Transaction	1..*
Loans	1..*	have	LoanType	1..1
	1..*	have	Banks	1..1
	1..*	have	Guarantor	[2]

2.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Members	member_id (PK)	int	Uniquely identifies each member	No	No	5
	name	varchar(255)	Members’ full name	No	No	Ahmad
	ic	int	Members' identification card number	No	No	981119-01-8732
	email	varchar(255)	Account email for receive notification	No	No	ahmad@gmail.com
	sex	varchar(255)	Members’ gender	No	No	Lelaki
	phone	varchar(255)	Members’ phone number	No	No	011-16596391
	DOB	date	Members’ birth date	No	No	19/11/1998
	noPF	int	Members’ personal profile	No	No	6
	address	text	Members’ address	No	No	No 128 Taman U
	status	ENUM	Indicates the application state	No	No	Johor
	city	text	City where the member is located	No	No	Muar
	postcode	int	Postcode where the member is located	No	No	84000
	state	text	State where the member is located	No	No	Kelantan
	office_address	text	Office address of member	No	No	Peti Surat 127
	office_city	text	Office city of member	No	No	Bandar Kota Bharu
	office_postcode	int	Office postcode of member	No	No	15710

	time	date	The date of submission application	No	No	05/1/2025
	familyNo	int	Family record of members	Yes	Yes	2
	staffNo	int	Staff number of member	No	No	4672
Admin	admin_id (PK)	int	Uniquely identifies each admin	No	No	1
	adminPassword	varchar(255)	Account password for admin login	No	No	$gf@142@
	adminUsername	varchar(255)	Admins' full name	No	No	Ahmad
	adminEmail	varchar(255)	Account email for admin login	No	No	admin@gmail.com
Transaction	Transaction_id (PK)	int	Uniquely identified transaction record	No	No	6
	transactionType	ENUM	Type of transaction made	No	Yes	Loan repay , Deposit
	time	date	Date of transaction made	No	No	19/1/2025
	Savings_id (FK)	int	Indicated to the specific savings account info	No	No	7
Guarantor	GuarantorNo (PK)	int	Uniquely identified guarantor of member	No	No	7
	name	varchar(255)	Guarantors' name	No	No	Siti
	ic	varchar(255)	Guarantors' IC	No	No	991231-01-9823
	staff_id	int	Guarantors' staff ID in KADA	No	No	7
	phone	varchar(255)	Guarantors' phone number	No	No	019-7113981
	application_id (PK)	int	Uniquely identified	No	No	6

Loans			each application record			
	date_apply	date	Submission date of application	No	No	20/1/2025
	loan_amount	float	Amount of loan to apply	No	No	2000.50
	interest_rate	float	Interest rate for loan	No	No	0.05
	monthly_repayment	float	Monthly repayment of member	No	No	200
	monthly_gloss_salary	float	Monthly gloss salary of member	No	No	4000
	monthly_net_salary	float	Monthly net salary of member	No	No	3500
	loan_period	date	Period of loan apply	No	No	18/2/2028
	status	ENUM	Status of the application	No	No	approved
	time	date	Submission date of the application	No	No	19/1/2025
	noPF	int	Indicated to the personal profile of member	No	No	4
	member_id (FK 1)	int	Indicated to member ID	No	No	8
	loan_type_id (FK 2)	int	indicated to type of loan member apply	No	No	4
	bank_id (FK 3)	int	Indicated to bank record of member	No	No	6
LoanTypes	loan_type_id (PK)	int	Uniquely identified the type of loan	No	No	5
	loan_type	varchar(255)	Type of loan to be applied	No	No	Simpanan Tetap
Banks	bank_id (PK)	int	Uniquely identified each members' bank info	No	No	8

	bank_name	varchar(255)	Members' bank name	No	No	Maybank
	bank_account	int	Members' bank account number	No	No	123456789000

2.4 Relational Database Schema

1. Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

   Primary Key: member_id

2. Admin (admin_id, adminPassword, adminUsername, adminEmail)
   Primary Key: admin_id

3. Transaction (transaction_id, transactionType, time, savings_id, loan_id)
   Primary Key: transaction_id
   Foreign Key: loan_id

4. Guarantor (GuarantorNo, name, ic, staff_id, phone)
   Primary Key: GuarantorNo

5. Loan (loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id)

   Primary Key: loan_id
   Foreign Key: member_id, GuarantorNo, loan_type_id, bank_id

6. Loan Types (loan_type_id, loan_type)
   Primary Key: loan_type_id

7. Banks (bank_id, bank_name, bank_account)
   Primary Key: bank_id

2.5 Normalization

Member Table

1. 1NF

MemberForm (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id, GuarantorNo, name, ic, staff_id, phone, loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id, loan_type_id, loan_type, bank_id, bank_name, bank_account, transaction_id, transactionType, time, savings_id, loan_id)

Primary Key: member_id

2. 2NF

Members (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

Primary Key: member_id

Loans (loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, loan_type_id, loan_type, bank_id, bank_name, bank_account, GuarantorNo, name, ic, staff_id, phone, transaction_id, transactionType, time, savings_id

Primary Key: loan_id

Foreign Key: member_id

3. 3NF

Members (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city,postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)

Primary Key: member_id

Loan (loan_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gloss_salary, monthly_net_salary, loan_period, status, time, noPF, member_id, GuarantorNo, loan_type_id, bank_id)

Primary Key: loan_id

Foreign Key: member_id, GuarantorNo, loan_type_id, bank_id

Loan Types (loan_type_id, loan_type)
Primary Key: loan_type_id

Banks (bank_id, bank_name, bank_account)
Primary Key: bank_id

Guarantor (GuarantorNo, name, ic, staff_id, phone)
Primary Key: GuarantorNo

Transaction (transaction_id, transactionType, time, savings_id, loan_id)
Primary Key: transaction_id
Foreign Key: loan_id

Admin Table

1. 1NF

AdminForm (admin_id, adminPassword, adminUsername, adminEmail)
Primary Key: admin_id

2. 2NF

The table is in 2NF because there are no partial dependencies. All non-key attributes are fully dependent on the primary key, admin_id. Therefore, 2NF is same as INF.

3. 3NF

The table is in 3NF because it satisfies 2NF, and there are no transitive dependencies as non-key attributes do not rely on other non-key attributes. Therefore, 3NF is same as INF.